



March 2026

Tutorial: Bathymetry-QGIS Module

Design: Roland (Ypsos)

1. Introduction

Module designed and specified by Roland (Ypsos) for Ortho4XP V3. It replaces the manual QGIS procedure (loading + extents spreadsheet + assembly) with a button-driven interface. Everything goes through rasterio, with no Terminal command whatsoever. Source files are never modified.

Overview

The Bathymetry module automatically prepares and assembles the bathymetry (the seabed) of an Ortho4XP tile, from depth surveys (EMODnet, SHOM/Litto3D...).

Logical order of the steps (shown at the top of the window):

Create structure → Load the data → Prepare · EPSG 4326 (if conversion needed) → Assemble

Fundamental principle: assembly selects files by their GEOGRAPHIC EXTENT, never by the folder name. Sorting by country is only for YOUR own organisation.

Bathymetry — Ortho4XP V3

Bathymetry — +46-003

Structure → Prepare the data (once per country) → Assemble the tile

Ready.

Tile +46-003 — extent -3.100 45.900 -1.900 47.100
 Sources: /Users/r.../top/Bathymétrie/Données EMODnet/France
 Output: /Users/r.../p/Bathymétrie/Assemblage tuile

1 source(s) found — from: EMODnet data + Reduced EPSG
 · E4_2024.tif

Click the Assemble button to start.

Overlap margin (%): (0.1 = 10% of the tile on all 4 sides) · Reference depth (m below sea level): (max 100)

QGIS: /Volumes/SSD/Utils_SCENES/QGIS/QGIS.app
 Sources: /Users/r.../p/Bathymétrie/Données EMODnet/France
 Output: /Users/r.../p/Bathymétrie/Assemblage tuile/+46-003

Prepare · EPSG 4326	Assemble tile	Clear Bathymetry sources	Refresh
EMODnet data	Bathymetry sources	Reduced EPSG	Tile assembly
Create structure	Add a country	Check (self-test)	Close
Free bathymetry (EMODnet)	Install the EMODnet file	Choose QGIS	Open in QGIS

The 4-folder structure

The module works inside a « Bathymetry » organisation containing four folders, each with its own button:

- Bathymetry sources / <Country> — your RAW data TO CONVERT (e.g. SHOM / Litto3D in Lambert-93; .asc, .tif, .xyz formats...). Only sources that are not already in EPSG:4326.
- Reduced EPSG / <Country> — the files CONVERTED to EPSG:4326 by « Prepare », ready to assemble. For bathymetry the resolution is NOT reduced (every seabed detail matters). Button shown: « Reduced EPSG ».
- EMODnet data / <Country> — the EMODnet files, ALREADY in EPSG:4326, read DIRECTLY by the assembly (no conversion). The free, fast source, to be preferred.
- Tile assembly / <tile number> — the final tile <tile>.tif, the one that feeds custom_bathy_dem.

Step-by-Step Workflow

1) Load the data (from the providers)

Get the bathymetry data from the official sources. Two main options:

- EMODnet (Europe): covers all European seas, ALREADY in EPSG:4326, in GeoTIFF format — open and free. Recommended source → « EMODnet data » folder.
- SHOM / Litto3D (France): very fine near the coast, but in Lambert-93 (.asc/.xyz) — to be converted → « Bathymetry sources » folder.
- GEBCO (worldwide): very coarse (~450 m), as a last resort only.
- Straddling tile (e.g. France + Belgium): simply drop in both datasets — the assembly merges them on its own.

EMODnet is already in EPSG:4326: no conversion, it goes to « EMODnet data ». SHOM/Litto3D is in Lambert-93: it goes to « Bathymetry sources », to be converted (Step 3).

Put each download in the relevant country folder, inside « Bathymetry sources » (see Step 2).



2) Open the module

In ORTHO4XP_V3 → Advanced menu → Bathymetry.

The window opens with the current tile as its title (example: +48+007).

[Screen: « Bathymetry — +48+007 » window. At the top, the reminder of the steps; in the centre the log (dark area); at the bottom the buttons.]

Click « Advanced ».



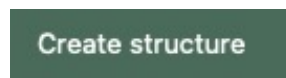
ou Click « Bathymetry ».



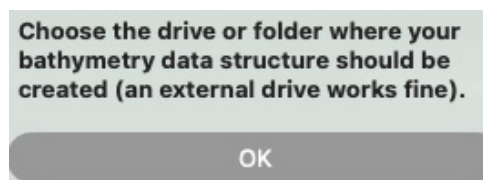
2) Set up the structure

1) Structure

Click « Create structure ».



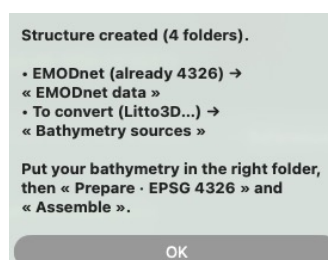
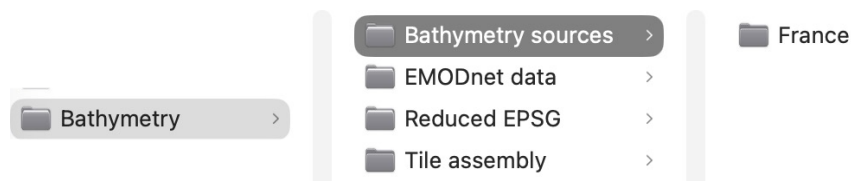
Choose the host disk or folder (an external disk is fine). Avoid Desktop and Documents (see the macOS warning below).



Enter the name of a first country (e.g. France).



The module creates all 4 folders at once, with the country subfolder inside the first 3.



For each additional country (Switzerland, Belgium, Germany...), click « Add a country » and enter its name. The subfolder is created in Bathymetry sources, Reduced EPSG and EMODnet data — never in Tile assembly (which is sorted by tile number).

Add a country

You may create your own sub-subfolders in Finder (e.g. Bathymetry sources / Belgium / Wallonia): the scan goes down recursively, so your organisation is free.

IMPORTANT: the folder buttons (Bathymetry sources, Reduced EPSG, EMODnet data, Tile assembly) only OPEN an existing folder. They never create the folder tree: only « Create structure » does.

3) Prepare the data

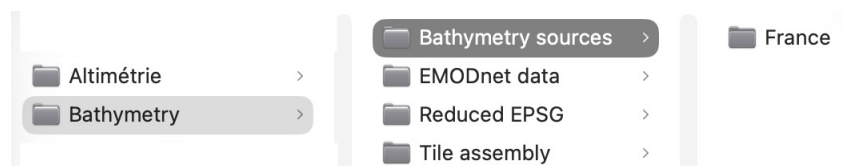
1) Prepare · EPSG 4326 (convert)

To be done ONLY for the sources that need converting (SHOM/Litto3D). Reproduces the QGIS chain gdalbuildvrt → gdalwarp (to EPSG:4326). EMODnet, already in 4326, does not need this step.

Click « Prepare · EPSG 4326 ».

Prepare · EPSG 4326

Choose the folder of raw data to prepare (it opens on Bathymetry sources): for example Bathymetry sources / France.



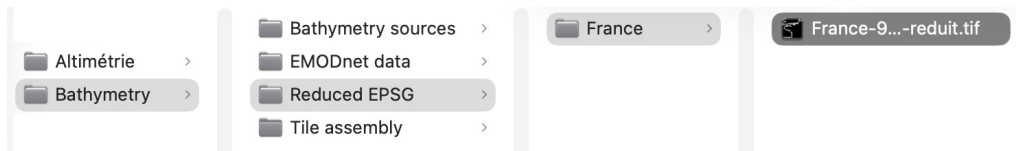
The module detects the source resolution (e.g. ~96 m for EMODnet, a few metres for Litto3D) and reprojects it to EPSG:4326.

- EMODnet (already 4326): does NOT go through « Prepare · EPSG 4326 », and stays in « EMODnet data ».
- SHOM/Litto3D (Lambert-93): goes through « Prepare · EPSG 4326 ». The module looks for the data in the Bathymetry sources folder and the EPSG:4326 result is written to « Reduced EPSG ».

Reduced EPSG is selected by default. Save the file there (e.g. France-reduit.tif).

Wait for the work to finish

Result: a file in EPSG:4326, stored in « Reduced EPSG », ready for assembly.



EMODnet note — do we need « Prepare »? NO: EMODnet is already in EPSG:4326, it goes straight to « EMODnet data » and the assembly reads it as is. « Prepare » is only for SHOM/Litto3D.

- EMODnet: use it AS IS, without « Prepare » — it is already in the right format and the right coordinate system.
- SHOM/Litto3D: run it through « Prepare » (reprojection Lambert-93 → EPSG:4326) before assembly.

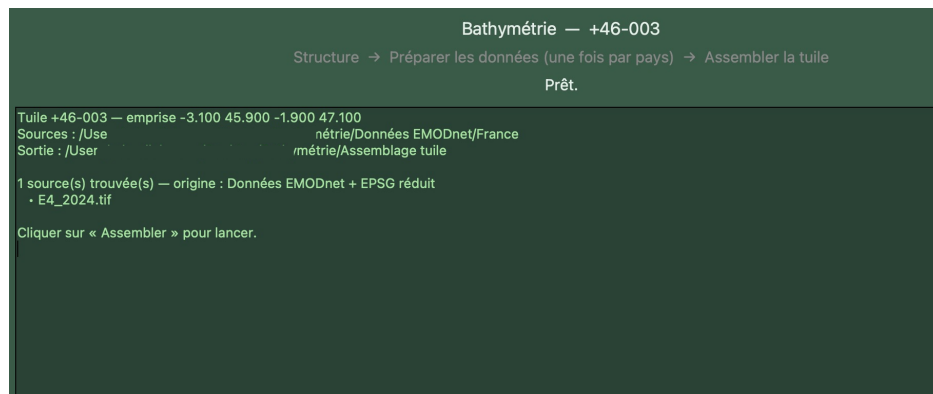
⚠ *The DEPTH values (e.g. -40, -89 m) are NEVER changed by « Prepare » or « Assemble »: only the projection changes. The negative sign (below sea level) is preserved.*

4) Assembling the data

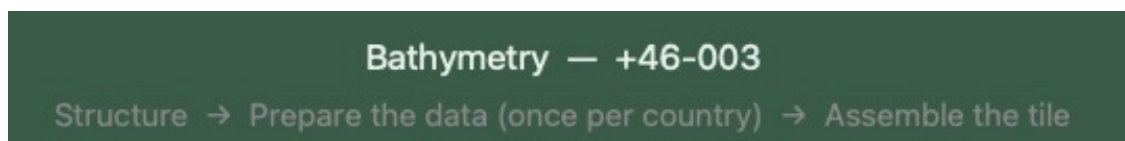
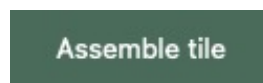
1) Assemble the tile

By default the active tile in ORTHO4XP_V3 is the one entered in the Latitude/Longitude interface (e.g. +48+7).

Optionally click « Refresh »: the log lists the sources found (kept / discarded + reason).

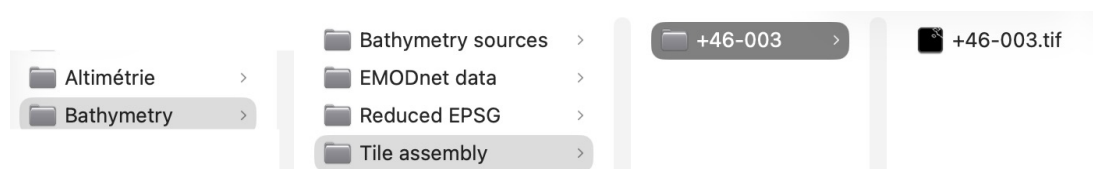


Click « Assemble ».



The module scans « EMODnet data » and « Reduced EPSG », across all countries, and keeps only the files that cover the tile enlarged by the overlap. It clips them, merges them and writes Tile assembly / <tile> / <tile>.tif, then sets custom_bathy_dem in the tile's .cfg.

Example: for a straddling tile, one EMODnet dataset on the France side and one on the Belgium side (EMODnet data) are selected and merged automatically, with no manual setting.



[Screen: the log lets you follow the steps « clip 4326 window ... / KEPT ... » for each source, then « Merging N source(s)... » and « Written: .../<tile>.tif ».]

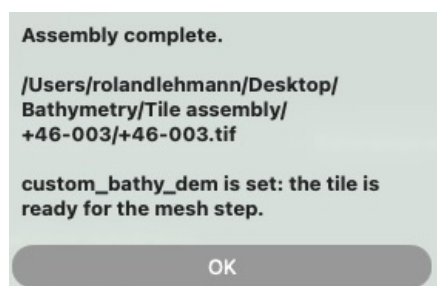
```
Bathymetry — +46-003
Structure → Prepare the data (once per country) → Assemble the tile
Done.

[DEM] Tuile +46-003 — emprise -3.100 45.900 -1.900 47.100 (débord 0.100°)
découpe fenêtre 4326 : 1152 x 1154 px
RETENU : France-97M-reduit.tif
[DEM] Fusion de 1 source(s)...
[DEM] Écrit : /Users/r... ktop/Bathymetry/Tile assembly/+46-003/+46-003.tif

custom_bathy_dem set in the tile cfg.
/Users/r... ilications/X-Plane_12/Custom Scenery/zOrtho4XP_+46-003/Ortho4XP_+46-003.cfg
Reference depth (bathy_max_depth): 100 m (below sea level)

DONE.
```

Operation finished



5) Very large country files

A file covering a whole country can be tens of GB (e.g. Germany 4M ≈ 148 GB). The module NEVER loads it whole: if it is already in EPSG:4326, it clips only the tile's WINDOW (reading in row blocks, constant memory) — the equivalent of a gdalwarp -te. Assembly stays fast whatever the size.

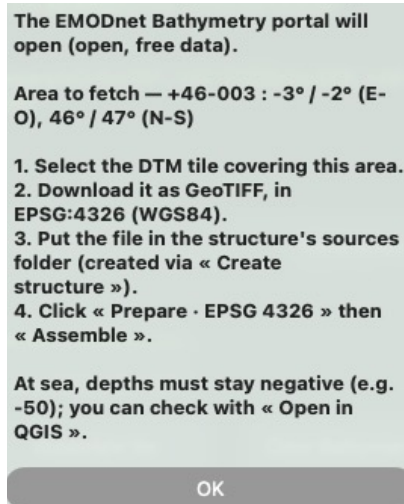
Tip: to avoid duplicating a 148 GB file, place a SYMBOLIC LINK to the real file in EMODnet data / <Country>. The module follows the link, even if its name does not end in .tif.

6) EMODnet bathymetry (free source)

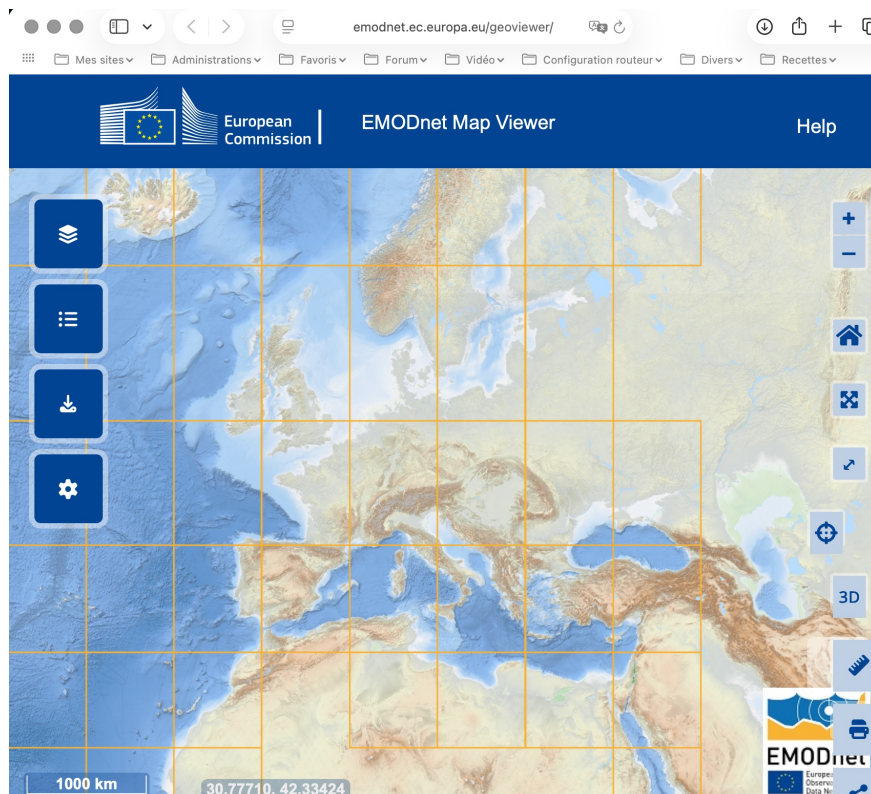
Button: Free bathymetry (EMODnet) / opens the portal at the following link:
<https://emodnet.ec.europa.eu/geoviewer/>

Free bathymetry (EMODnet)

Click OK



Zoom in or out on the map to display the rectangles that delimit the areas.



Click in the centre of the desired area



In this window select the line: 2024 E4, 32-bit float GeoTIFF and click Download



European
Commission

EMODnet Map Viewer

Help

Tile structure and download

DTM Release	DTM Tile	Product format	Download URL
☐ 2024	E4	32 bit float GeoTiff	Download 
☐ 2024	E4	EMO	Download 
☐ 2024	E4	EMO (without GEBCO data)	Download 
☐ 2024	E4	ESRI ASCII	Download 

EMODnet is ~96 m: coarse but enough for the water rendering.

For more detail near the coast, use SHOM/Litto3D (via « Prepare »). Two cases:

- EMODnet (~96 m): usable as is for the water rendering, without « Prepare · EPSG 4326 ».
- Litto3D (Lambert-93): run it through « Prepare · EPSG 4326 » to reproject it to EPSG:4326 before assembly.

The goal: a clean bathymetry in EPSG:4326 for the tile:

- EMODnet for the coverage,
- SHOM/Litto3D for the detail near the coast.

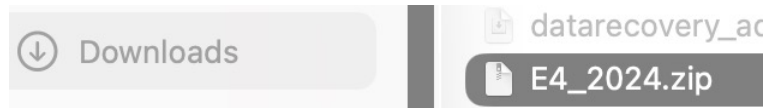
The file is in Downloads

7) Install the EMODnet file

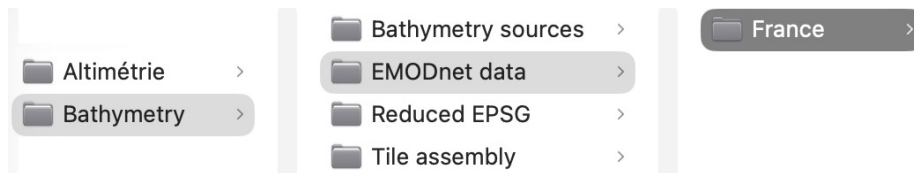
Click the « Install the EMODnet file » button.

Install the EMODnet file

In Downloads, select the downloaded file and click Open



It will unzip the downloaded file (ZIP or GeoTIFF) and place it directly in « EMODnet data » / <Country>. No separate disk to set.



8) Clear Bathymetry sources (cleanup)

Button: Clear Bathymetry sources

Clear Bathymetry sources

Once your sources are converted, you can free up space. The button shows a CHECKLIST of the files present in « Bathymetry sources », with an indicator:  already converted (present in « Reduced EPSG ») /  not yet converted (keep it).

Triple safety: nothing is deleted without (1) your tick, (2) an alert if a non-converted file is ticked, (3) a final confirmation. Never automatic, never in bulk.

9) Permanent settings

Overlap margin (°)

Overlap margin (°):	0.1	(0.1 = 10% of the tile on all 4 sides)	·	Reference depth (m below sea level):	100
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Default value 0.1 (= 10% of the 1° tile on all 4 sides). Ensures the edges of neighbouring tiles overlap seamlessly. Do not change unless you have a specific need.

Reference depth (m below sea level)

« Reference depth (m below sea level) » field, default 100, CAPPED at 100. It is the depth that maps to the « deepest » water in the rendering (bathy_max_depth key in the .cfg). Over a shallow coastal area (e.g. Ile d'Yeu, depths from -34 to -89 m), set 50 or 90 so that the water colour gradient matches your real depths. This value NEVER changes the data: it is only the normalisation reference. It is written to the tile's .cfg when assembling.

QGIS / Choose QGIS / Open in QGIS

Choose QGIS

Optional path to the QGIS application, to open the assembled <tile>.tif and check the depths (negative values = below sea level).

10) Result and checking

The <tile>.tif file is written to Tile assembly / <tile> and pointed to automatically by custom_bathy_dem. Check in QGIS that the tile is FULL (VALID_PERCENT = 100 statistic, no holes), then run the build in X-Plane 12 to check the result in flight.

Note: the produced file is compressed (DEFLATE) — it is therefore lighter than an uncompressed QGIS export, with strictly identical data.

macOS warning — permissions

If the log shows « Operation not permitted » on a file, this is a macOS privacy protection: your structure is probably on the Desktop or in Documents. Fixes: allow the app in System Settings → Privacy & Security → Full Disk Access, OR place the Bathymetry structure outside Desktop / Documents.